Reflective Journal

Benjamin LaCount II | ITAI 2373 | October 17, 2026 | Solo project

https://github.com/MarineMaaan/ITAI2373-NewsBot-Midterm

Independent work and preparation

I completed this project as a solo assignment. There was no group name or number, and no video demonstration was included. In place of team role allocation, the work required connecting planning, implementation, analysis and documentation into one coherent deliverable. This made consistency especially important: a decision about the dataset or preprocessing had consequences throughout the notebook.

I took more time than most to prepare for the assignment. That preparation helped clarify the requirements, but it also showed me that I need a more efficient way to begin. In future projects, I want to turn the instructions into a short requirements checklist, confirm the data format early, and establish a small working pipeline before expanding the analysis. Preparation should reduce uncertainty without delaying the first useful result.

Learning from preprocessing

My strongest lesson was how much preprocessing influences the quality of the final output. Cleaning text is a modeling decision, not merely a preliminary chore. Removing punctuation, capitalization, stop words or numbers can simplify a classifier while also removing information needed by another NLP task.

The NewsBot design addresses that tension by preserving normalized article text for sentiment, parsing and entity recognition, while using tokenized, lemmatized content for TF-IDF classification. The BBC source is largely lowercased, so preserving its text cannot recover capitalization that is already missing. Recognizing that limit is part of interpreting the results responsibly.

Technical integration challenges

The central integration challenge was making several NLP techniques work together without giving them identical inputs or overstating what they measure. A category prediction, a sentiment score and an extracted relationship answer different questions. An entity co-occurrence edge shows that two names appeared together; it does not establish a business relationship.

Evaluation creates another important boundary. Vocabulary learning and feature selection belong inside training cross-validation, while reserved test articles measure generalization. Duplicate removal must happen before splitting. These choices connect data preparation to the credibility of the final model comparison, and they provide a more useful lesson than a high accuracy score alone.

From news analysis to studio insight

Business value assessment

The application I care about is reading and categorizing game reviews for products my studio produces. A review-monitoring system could help identify recurring concerns about bugs, performance, gameplay and value, then direct attention to the most useful feedback. The practical benefit would be a clearer view of player concerns and a more efficient review process.

NewsBot is a foundation for that idea rather than proof that it already solves it. Its BBC labels describe news topics, not review themes. Whole-article sentiment can also miss whether a reviewer likes a game but dislikes one feature. A studio version would need appropriately sourced review data, new labels and evaluation across different games and review sources.

Individual contribution and AI assistance

I used AI to assist with executing the work while preserving the thought, planning and final quality assurance as my own responsibilities. That assistance included implementation, debugging, evaluation support and drafting documentation. I supplied the personal reflection themes and the studio use case. I remain responsible for understanding the pipeline, checking the outputs and making the final submission decisions.

Working solo does not remove the need for accountability. The repository makes the process inspectable through the notebook, reusable code, evaluation outputs and written explanations. Clear attribution of assistance matters because the project should demonstrate understanding and judgment as well as the ability to produce a working artifact.

Future enhancements and professional development

My next steps would be to prepare more efficiently, build a small review annotation set, and compare useful review categories before adding more model complexity. Human review of uncertain predictions should remain part of the workflow. Entity extraction and emotion analysis also need their own labeled evaluation data rather than relying on the topic classifier's results.

The bonus experiments broaden the project through an interactive dashboard, a controlled bilingual comparison, custom domain NER training and a measured feature ablation. Their small or synthetic evaluation sets have limits. For professional development, the key habit is to state those limits clearly, connect results to a real user need, and make the evidence reproducible. That approach is directly relevant to building analytical tools for my studio.